

Runtime

Device: CPU

Checkpoint path



/mnt/c/Users/jaswa/Downloads/net_39

Default:

/mnt/c/Users/jaswa/Downloads/net_39epoch.
h.pth

Loaded: net_39epoch.pth

SmartSpectra — RGB to Hyperspectral Reconstruction

Model 1 (fine-tuned MST++) — reconstructs a **31-band** hyperspectral cube (400–1000 nm) from a single RGB photo. Output is a *reconstruction estimate*, not a measured spectrum.

Upload a fruit photo



appl.png
49.5KB



Input RGB

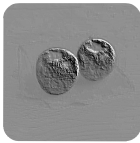


Reconstructed cube: 900×900×31 (float32, raw range -0.0218 ... 0.0308, mean 0.00000)

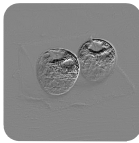


The checkpoint's raw output magnitude is much smaller than the [0, 1] reflectance scale it was trained to predict. Band images below use a percentile-based contrast stretch (1st–99th percentile → [0, 1]) purely for visibility. Treat them as relative intensity, not quantitative reflectance.

Reconstructed spectral bands (sample slices)



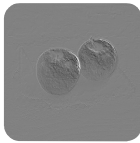
~400 nm
(band 0)



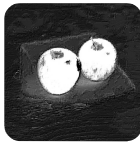
~540 nm
(band 7)



~700 nm
(band 15)

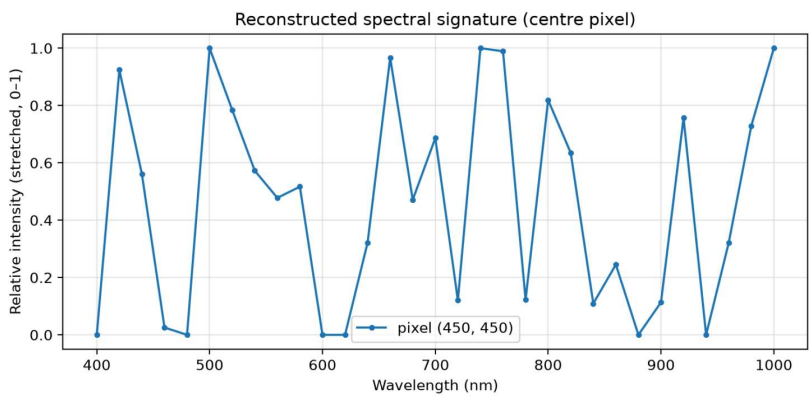


~860 nm
(band 23)



~1000 nm
(band 30)

Spectral signature at the centre pixel



> Pick a different pixel



This shows **Model 1** (RGB→HSI reconstruction) only. **Model 2** (pesticide-risk classification on top of Model 1) is not integrated into this demo yet.

Note on accuracy: the model is fine-tuned on Agro-HSR at 400–1000 nm, but the input is a 3-channel RGB photo — a physically limited signal. Reconstructed spectra are estimates, not measured values. Treat them as approximate indicators of spectral shape, not as quantitative truth.